# Io, gli Elefanti e la Teoria dei Giochi

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Formal Models in Comparative Politics & International Relations

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## Abstract

Standard game-theoretic accounts often rationalize departures from fully rational play by appealing to exogenous "shocks" or moments of surprise. Yet many strategic encounters suggest that what matters is not merely an external disturbance, but an endogenous cognitive-emotional shift that reshapes perception, signaling, and restraint. This paper develops an illustrative framework for endogenizing surprise and empathy in signaling settings, using two elephant-human encounters reported by Safina (2015) as motivating cases. In one episode an elephant charges a tourist yet brakes before contact; in another an elephant injures a shepherd and subsequently guards him until help arrives. Canonical signaling models can capture incentives and credible threats in both interactions, but they struggle to accommodate the joint pattern "attack, then care." I treat the two episodes as potential outcomes of a common underlying interaction and use a causal directed acyclic graph to make explicit the missing variables that standard models implicitly omit. I argue that surprise operates as an informational shock that can trigger aggression, while empathy functions as an affective regulator that modulates signal intensity and can generate post-conflict assistance. To formalize this intuition, I sketch an intrapersonal adaptation of Groseclose and Snyder's (1996) vote-buying game, in which competing emotions allocate limited intensity across thoughts to secure behavioral dominance. The resulting perspective offers a tractable route to richer models of bounded rationality and yields testable implications for when shocks produce escalation versus restraint.

# 1 Introduction

In this paper I wish to describe an intuition regarding the way game thoery has yet formalized some interactions between agents. Game thoerists consider surprises and shocks as the rationale for an irrational behaviour. I believe that the following paragraphs will convice the audience about the value of this contribution in two sense; first, I hope the description provided will be further used and applied. Second, the intuition about the role of emotions adds a new perspective to the literature about bounded rationality.

The intuition regards the field of animal behaivour, that has drown a lot from game thoery. Even though it may appear far from what economists study, still, it deals with the problems that game thoerists often face: the first is to represent the processes that happen in agents mind, this means taking into consideration and "measuring" variables that we do not observe. The second problem deals with the creation of a formal idea about this processes. The last problem connects this idea with the observed behaviour, and estabilishes an equilibrioum, thus a state of the world with particular properties. I aim to show that seeking an explanation outside the mind is unnecessary, since it may well lie within it.

In order to ease the description and to make the models closer to a possible application, I expolit a singular situation described in Safina (2015). This book is an amazing challenge for every person who has ever had to face a game thoery excercise. To support his thesis, the author provides the reader detailed descriptions of animal interactions. All species selected belong to the realm of "cleaver" animals, those animals capable of emotions and complex cognitive processes. I believe that each of these interactions is a good game theory excercise, since they forces the game thoerist to apply its knowledge out of his area of expertise.

This book also provides a simplified setting for these exercises, since animals constitute the perfect agents in game thoery. They are complex enough for beign represented as rational players, but they are not as complex as humans, so that most of the assumptions and simplifications, have more grasp into reality. Under this respect, elephants appear to be the perfect application. There is plenty of evidence about their behaviours, and there is plenty of evidence about the fact they act in accordance with a *mind*. According to Safina (2015), they are almost as humans, and they often interact with humans.

I will formalise my intuition thorugh a Causal Directed Acyclic Graph (DAG). Since I wish to norrow down the gap between thoery and practice, I believe that this tool perfectly suits my research design. Additionally, provide a more transparent version of the analysis, clarifiyng the variables of interests and the limits of the assumptions.

The final aim of this article is to clarify a contradiction between two similar situations that game theory formalises in different ways.

## 1.1 Beyond Words

Carl Safina is a biologist and wrote a world-famous essay (Safina, 2015) about animals' minds. He argues that until now, biologists have refused to deepen the study of animals' minds and "souls" and that is high time they did that. The curious thing about this book is that, in doing so, he tries to bring to readers' attention a problem quite close to game theory. He tries to rationalise the behaviour of some animals, arguing that their behaviours may be explained if we accept to consider them "intelligent", thus guided by a mind and sensitive to emotions. This book is a tentative to take animals mind at the same level of human minds. I hope the animal-world examples will:

- change the way we theorise a broader set of agents and interactions.
- be tested to reinforce the empirical evidence presented by Safina.
- provide a new equilibrioum concept.

## 1.2 The Problem

This section aims at claryfing the setting to which the game thoery models refere, as well as making the reader aware of the basic intuition. In the examples Safina describes in his book (Safina, 2015) animals seem to be driven by rationality and emotions, as much as humans are. *These are brief summaries of the two episodes.* Both interactions between humans and elephants happened in a wildlife park in Africa. In both, the surrounding environment appears to be similar.

**Case 1:**

A staff member of a wildlife park in Africa reports an "attack" by an elephant on a tourist. The scene described clearly highlights the signals that the elephant sends to the tourist. It also clearly identifies the reasons why it is logical to believe that the elephants intentionally avoided hurting the tourist. In particular the staff member reports signs on the ground that undoubtedly mark the intention to halt the charge against the tourist moments before impact. We can formalise this game through a signalling game in which the elephant sends a signal to threaten the tourist but doesn't have any intention of hurting him.

Another situation, recorded in the same environment, shows how similar conditions can produce an opposite pattern of behaviour.

**Case 2:**

In the same wildlife park in Africa, a rescue group was involved in a mission to reach a shepherd who was badly hurt by an elephant. When the staff approached him, they reported that the shepherd stopped them from harming the animal. He argued the elephant had kept him safe during the night. From the behaviour of the animal it appeared evident what was going on in his mind. After the attack, the elephant soon realised it had misinterpreted the signals from the shepherd, and after hurting him, the animal took care of him until the rescue group reached the place of the accident. Here, the same signalling structure of the first example breaks down: the equilibrium is reached but the elephant attacks. However, the event suggests a richer model of rationality, where the animal is able to recognise an "error".

Both scenarios can be framed within game theory, yet each reveals elements of cognition and emotion that standard models overlook.

In particlar, we shall take into consideration three contradictions that emerge from these examples:

1. Why does the animal attacks and then helps? Isn't it contradictory?
2. Why does the animal attacks only in one of the two cases?
3. What is rational? To attack and regret that decision or not attack at all? With respect to what criteria do we define which is the best strategy?
4. Which is the beste strategy in evolutionary terms?

## 1.3 Parallel with Game Theory

I believe game theory analysis accurately represents most of the aspects of both cases. For example, the fact that the animal starts a conflict with humans and tends to be pacific with most of other creatures is due to the fact that humans are one of the few threats they face in nature. Game theory correctly models agents as driven by incentives, it does not take into account interactions in which the agent acts violently because *it has a bad nature*. On the same line, game theory models credible threats as equilibria depending on the information available to players: not the turist nor the sheperd intended to porpusly harm an animal that may easly kill them, for example.

Additionally, there are plenty of models that suite each of the two situations. For instance, as we will see, some models explain the behaviuor of the first case thorugh the concept of mimic: the elphant knows he is not going to do any harm to the tourist. However, the turist cannot distinguish between an animal with good or bad intentions, and the elephant exploits this weakeness for defense porposes.

Thus, both animals and humans may be depicted as rational agents, since their behaviour appears to be consistent with the environment and with a broader goal, if not with an aim in mind. Both appear to be able to communicate with each other, indicating that both species are adapting to signals from others and that they can act as informed agents. And both seem to be aware of each other's strengths, at least partially (Schelling, 1960; Smith, 1973). Again, in both cases, the animal seems "pacific" and willing to avoid or prevent conflict, at least at a certain point in time.

## 1.4 Representative Agents

In both cases analogies between men and animals' minds do not appear to be crystal clear. However, most of their differences need not being represented in game theory formalisations. Additionally, game theory is often used to study nations behaviours, this means that only the crucial features of agents need to be represented.

Thus, under the descriptive limits of these games and under their assumptions, I believe no harm is caused when I use the terms human, elephant and agent as synonyms. *I will differentiate among these agents only in those cases in which it is necessary, and with due advertisement to the reader.*

# 2 Causal Inference

The discussion so far motivates a comparison with frameworks that model how hidden factors influence observed outcomes. Causal inference (Pearl, 2009) suggests that in order to answer this question we may also think of an unobserved confounder that we are mistakenly excluding from the framework. I argue that in this context this missing variable exists.

## 2.1 Application of Potential Outcome Framework

This section formalises the intuition using a DAG. To make the subsequent discussion accessible, it first outlines how the potential-outcomes framework applies to this setting.

A potential outcome is a counterfactual. Thus, it is a scenario that we do not observe; we think of it as being the closer and most realistic scenario that might have happened in case the actual one didn't.

In the case of the two intaractions I presented, each of them may be considered the counterfactual scenario of the other. Since counterfactuals do not exist, we use real and observed examples to mimic them. The underlieng belief, is that no researcher is as good as reality itself in producing examples of what might have happened. If we trust reality and observation of reality, the role of the researcher is limited to selecting the most appropriate counterfactual from those offered by natural world.

*Hypothesis:* under no effect of emotions, the case offered by the elephant and the sheperd would have been identical to the one offerd by the other elephant and the tourist.

## 2.2 The Reason why I want to use Causal Inference
### From the General Case to the Single Case and Back Again

According to Von Neumann and Morgenstern (1990, chap. 2, p. 14): "An almost exact theory of a gas, containing about $10^{25}$ freely moving particles, is incomparably easier than that of the solar system, made up of 9 major bodies; and still more than that of a multiple star of three or four objects of about the same size. This is, of course, due to the excellent possibility of applying the laws of statistics and probabilities in the first case. [...] The problem must be formulated, solved and understood for small numbers of participants before anything can be proved about the changes of its character in any limiting case of large numbers".

To pursue this suggestion, we go from the abstract general case of rational-agent model, to analyse specific interactions in which that model fails to adequately map reality.

We then turn to causal inference in order to reconstruct a more general account: namely, to move from the idiosyncrasies of a single interaction to a *new* general framework. This requires abstracting away from irrelevant features and identifying a shared causal mechanism. The result is a revised pattern, a new map that can guide inference, organize the evidence, capture the data-generating process.

# 3 Models

Let's repeat the three contradictions that emerged from the animal examples in subsection 1.2 (The Problem):

1. Why does the animal attacks and then helps? Isn't it contradictory?
2. Why does the animal attacks only in one of the two cases?
3. What is rational? To attack and regret that decision or not attack at all? With respect to what criteria do we define which is the best strategy?
4. Which is the beste strategy in evolutionary terms?

## 3.1 Model 1 - Surprise as Shock

The decisive variable in this situation may be the element of surprise (Schelling, 1960). The two versions of the elephant-human encounter, the one involving the shepherd and the one involving the tourist, are analogous. In the tourist scenario, there is no surprise: the elephant is aware that humans might approach and disturb it, and it knows where to find them.

The situation involving the shepherd, however, contains the element of surprise, which may justify a different behavioral response, namely, the attack. What surprise alone does not explain, however, is why the elephant protected the shepherd after the attack.

To justify this behaviour we introduce empathy, force that shapes the outcome of the game.

However, before jumping to the model, we must ourselves what does empathy have to do with the first case. We may argue that empathy weakens elephant's signal. Yet empathy also functions as a force that regulates the intensity of the signal itself.

Is it therefore reasonable to suggest that, in the second case, empathy may also be the mechanism that modulates the variability of the signal? What is the relationship between these two forces?

Now that the cause has been established, we can turn to the effects. We observe two primary effects: that of surprise and that of empathy.

## 3.2 Model 2 - Vote Buying

### 3.2.1 Back to Groseclose and Snyder (1996)

In this section I propose a readaptation of a famous model from Groseclose and Snyder (1996). I intend to exploit the genious equilibrioum authors found, to support the intuition coming from the DAG with another nerrative. In doing so, I believe we are improving the previous model (See subsection 3.1: Model 1 - Surprise as Shock).

In their article, Groseclose and Snyder (1996) explain why in the US Congress the number of votes often exceedes the minimum required to make a bill pass. Thay argued that this is due to the interaction between lobbying groups competing for support of different bills. These groups *bribe* Congressmen to convince them.

The basic intuition from the paper comes from this example:

*"The legislature contains seven members, all indifferent between the two proposals, that is, $v(i) = 0$ for all $i$. Since $B$ will move second and attack the weakest part of $A$'s coalition, $A$ will offer the same bribe, $a$, to all legislators he bribes. Let $m + 4 \geq 0$ be the size of the coalition $A$ bribes. If $m = 0$, then $B$ can defeat $x$ by paying $a + \varepsilon$ to one member of $A$'s coalition. Thus, $A$ must set $a \geq W_B$ to win, and $A$'s total bribes are then $4W_B$. If $m = 1$, then $B$ must pay $a + \varepsilon$ to two members of $A$'s coalition to defeat $x$. Thus, $A$ must set $a \geq W_B/2$ to win, and his total bribes are then $5W_B/2$. This is less than $4W_B$, so $A$ is better off buying a supermajority of size 5 than a minimal winning coalition. If $m = 2$, then $B$ must pay $a + \varepsilon$ to three members of $A$'s coalition, so $A$ must set $a \geq W_B/3$ to win, and his total bribes are then $2W_B$. In fact, $A$ minimizes his total payments by bribing all seven legislators. Here, $m = 3$, and $A$'s total bribes are $7W_B/4$. Clearly, there is an equilibrium where $x$ wins if and only if $W_A \geq 7W_B/4$. If $W_A < 7W_B/4$, then there are only equilibria in which no bribes are paid and $s$ wins."* (Groseclose & Snyder, 1996)

### 3.2.2 Representation through Vote Buying

Consider a council composed of several members. These members are not strategic players in the game but rather represent passive agents whose votes can be influenced. There are two active players, conceptualized as two groups, each seeking to "buy" the votes of the council members in order to determine the policy agenda. Each group possesses a budget, and bills are voted on by the council according to a predefined rule, assumed here to be the simple majority rule. This setup corresponds to the classical *vote-buying game* as formalized by (Groseclose & Snyder, 1996).

The objective of this analysis is to adapt this framework to the so-called *elephant-shaped* case. This adaptation allows the equilibrium structure of the original model to be employed within a cognitive and emotional context.

- Let $K$ denote the total number of *thoughts*.
- The two players represent *opposing emotions* that regulate the internal dynamics of the elephant's mind. "Setting the agenda" refers to determining the behaviour.
- Each player's *budget* corresponds to the *intensity* of its emotion. A higher emotional intensity enables the emotion to "purchase" a larger number of thoughts.

Notation:

Players: $N = \{X, Y\}$ or $N = \{Emotion1, Emotion2\}$.

Histories: $H = \{\varnothing, \{x\}, \{x, y\}\}$. Where $x$ and $y$ indicate vectors of payments in the standard setting. Emotions use a unit of payment (emotional intensity) to corrupt a thought (or a judgement).

Player function: $p(\varnothing) = X; \quad p(x) = Y$

Actions: $A(\varnothing) = \{x\}; \quad A(x) = \{y\} \; \forall\, x$.

Each player *receives* $V_i$ if elephant acts accordingly.

Each player loses $\sum_{j=1}^{k} x_j$ for each unit of intensity spent.

This adapted framework enables a formal representation of conflict as a strategic interaction between emotions competing for cognitive control. Each emotion allocates its limited intensity across thoughts, attempting to secure a majority sufficient to determine the elephant's mental and behavioral trajectory. The resulting equilibrium can be interpreted as a stable emotional configuration in which neither emotion can improve its outcome given the other's allocation strategy. Let $\mu$ denote the minimal number of thoughts required to *set the agenda*, that is, to achieve behavioral dominance of one emotion over the other. Then:

$$\mu = \frac{k+1}{2}$$

# 4 Literature Review and Limitations

## 4.1 A review of models of conflicts

Further exploration of the literature on games of conflict may support as well as weaken the analysis presented so far. In any case, already-established models are a wonderful comparison. Some of the papers that stand out are (Schelling, 1960).

I think it may be of interest to focus on McKelvey and Palfrey (1995). A quantile response equilibria may be an interesting development of the model.

Modelli di cherim: Reiter (2003) says: "James Fearon prominently contributed to this scholarship with his 1995 paper "Rationalist Explanations for War." He usefully highlighted the point that if states agreed on the outcome of a possible war, they could probably avoid war. Central to Fearons paper is the importance of focusing on states' disagreement over their capabilities and/or resolve."

From Fearon: (Fearon, 1995, 1996, 1998)

### 4.2 Limits of Models

In many courses I covered the topic of formal modelling. In the previous pages I present my personal reelaboration along with a brief summary of all litarature I studied on this topic. This is for sure not enough to assess the state of models of this article. However, the exposition I provide the reader with, should be considered satisfactory in the sense that there will never be any such thing as *the correct, perfect or true modelling theory.*

Under this respect, it is possible to further expand the discussion on limitations. The literature offers some very good papers, among others, I report: Ashworth et al. (2021), Bräuninger and Swalve (2020), and Pareto (1916). The present different ways to combine empirical and theoretical researc along with some very interersting interpretations of the meaning and scope of models.

### 4.3 Connection between Game Theory and Biology

There are many great applications of game theory to biology. I believe the most interesting ones are: Axelrod and Hamilton (1981) and Smith (1973). Future works may shed light on more recent, as well as interesting applications. A vaste literature covers the topic of evolutionary games too: Hofbauer and Sigmund (2003), Kandori et al. (1993), Nowak (2006), and Taylor and Jonker (1978). In particular, Hofbauer and Sigmund (2003, chap 12) offers a clear treaty of powerful analytical tools along with applications to these game theoretic tools to biology.

Unfortunately, an extension dedicated to these numerous works must be postponed.

## Conclusion

The puzzle motivating this paper was deliberately simple: two elephant-human encounters occur in broadly similar environments, yet they generate sharply different behavioral trajectories. In one case the elephant's threat is forceful but calibrated; in the other, violence is followed by a striking form of care.

Standard signaling logic goes a long way toward explaining why an elephant might issue a credible threat and why a human might respond cautiously. What it does not readily explain is the combination of escalation and subsequent protection-an outcome that looks, from the standpoint of canonical rational-agent theory. The central claim advanced here is that the apparent inconsistency is largely an artifact of what the model leaves unmodeled: the relevant "shock" can be internal, and the relevant constraint can be affective.

The causal-inference lens is useful. Treating the two episodes as potential outcomes of a shared underlying interaction makes the research design explicit: the cases serve as mutual counterfactuals, and the goal is to isolate which latent factor-held implicit in standard game-theoretic-can plausibly generate the divergence. Within this framing, surprise functions as an informational discontinuity that plausibly shifts the elephant's perception of threat and thus the equilibrium-relevant mapping from signals to actions. Crucially, however, surprise alone cannot account for the post-attack behavior. That second step motivates the introduction of empathy as a regulator: a force that can attenuate aggression, reshape the "intensity" of signals, and generate assistance once the situation is reinterpreted.

This conceptual move has a modeling payoff. Instead of treating emotions as noise corrupting otherwise coherent strategic choice, the paper sketches a way to represent emotions as structured, competing influences over cognition. The adaptation of the vote-buying framework provides a particularly transparent metaphor: thoughts play the role of passive "voters", while emotions act as active players allocating scarce intensity to secure agenda-setting power-here, the determination of behavior.

In equilibrium, the elephant's observable trajectory corresponds to a stable emotional configuration: neither emotion can improve its outcome given the other's allocation. Read this way, "attack, then care" is not a contradiction so much as the observable trace of a reallocation of cognitive control after misinterpretation is recognized.

The broader implication is that many phenomena labeled as "shocks" in political science and international relations may be mischaracterized when treated as purely exogenous disturbances. A framework that makes these mechanisms explicit can therefore complement familiar strategic models, not by abandoning rationality, but by widening the set of state variables that rational choice is allowed to condition on.

At the same time, the limits of the present contribution should be stated plainly. A fuller treatment would require specifying payoffs, information structures, and equilibrium selection more rigorously. Future work could therefore proceed tightening the formal model, including the relationship between surprise and empathy as distinct but interacting state variables.

The take-home message is modest: when game theory explains puzzling behavior by invoking "surprise", some times it is pointing toward a missing internal mechanism. Making that mechanism explicit-rather than treating it as residual noise-can turn an ad hoc rationalization into a tractable extension of formal modeling, one that is better aligned with the texture of real strategic behavior.

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